

A Project Report
On
Developing Predictive Models for Early ASD Detection
Based on fMRI Scans

Submitted to

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY ANANTAPUR, ANANTHAPURAMU

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Submitted By

E. REDDY ROHITH	22691A05I1
J. REDDY MANOJ	22691A05I0
M. MONIKA	22691A05D4
J. MOUNIKA	22691A05D6
M. THRINESH NAIK	21691A05N6

Under the guidance of

Dr. R. Nidhya Ph.D.,

Professor

Department of Computer Science & Engineering



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This is to certify that the project work entitled “**Developing Predictive Models for Early ASD Detection Based on fMRI Scans**” is a bonafide work carried out by

E. REDDY ROHITH	-	(22691A05I1)
J. REDDY MANOJ	-	(22691A05I0)
M. MONIKA	-	(22691A05D4)
J. MOUNIKA	-	(22691A05D6)
M. THRINESH NAIK	-	(21691A05N6)

Submitted in partial fulfillment of the requirements for the award of degree **Bachelor of Technology** in the stream of **Computer Science & Engineering** in **Madanapalle Institute of Technology & Science, Madanapalle**, affiliated to **Jawaharlal Nehru Technological University Anantapur, Ananthapuramu** during the academic year 2025-2026.

Guide
Dr. R. Nidhya, Ph.D.,
Professor,
Department of CSE.

Head of the Department
Dr. M. Sreedevi, Ph.D.,
Professor & Head,
Department of CSE.

Submitted for the University examination held on:

Internal Examiner
Date:

External Examiner
Date:

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outstanding remarks at ELiteTech Intern.

Adarsh

Founder(ELiteTech)

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29-March-2026

ETI-AO-11709

Jonna Mounika

for successfully completing a 12 Weeks internship from 26-
December-2025 to 25-March-2026 in Java Developer With
outstanding remarks at ELiteTech Intern.

Adarsh

Founder(ELiteTech)

 elitetechintern.com



DECLARATION

We hereby declare that the results embodied in this project “**Developing Predictive Models for Early ASD Detection Based on fMRI Scans**” by us under the guidance of **Dr. R. Nidhya, Ph.D., Professor, Dept. of CSE** in partial fulfilment of the award of **Bachelor of Technology in Computer Science & Engineering** from **Jawaharlal Nehru Technological University Anantapur, Ananthapuramu** and we have not submitted the same to any other University/Institute for award of any other degree.

Date:

Place:

PROJECT ASSOCIATES

E. REDDY ROHITH

J. REDDY MANOJ

M. MONIKA

J. MOUNIKA

M. THRINESH NAIK

I certify that above statement made by the students is correct to the best of my knowledge.

Date:

Guide

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ABSTRACT

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition that affects social interaction, communication abilities, and behavioral patterns. Early and reliable diagnosis remains challenging because existing methods are largely dependent on expert-driven behavioral assessments, which can be time-consuming and subjective. In recent years, resting-state Functional Magnetic Resonance Imaging (rs-fMRI) has gained attention as a potential tool for identifying objective brain-based indicators of ASD. However, its application is often constrained by issues such as high data dimensionality, differences in data acquisition across sites, and motion-related disturbances, especially in younger individuals.

In contrast to conventional approaches that eliminate such disturbances during preprocessing, this work examines their potential significance as useful predictive signals. Scan-quality measures, including Framewise Displacement (FD), DVARS, and Signal-to-Noise Ratio (SNR), are integrated with phenotypic information obtained from the ABIDE-I dataset. To handle the complexity of the data, a Mutual Information-based feature selection method is employed to retain the most relevant attributes. These features are then used to train an optimized XGBoost classification model.

The proposed system is evaluated on a dataset comprising 1,112 subjects, including both ASD and typically developing individuals. The model achieves an accuracy of 95.96%, along with high precision, recall, and AUC-ROC values, indicating strong classification performance. The findings demonstrate that incorporating scan-quality metrics alongside phenotypic features can significantly enhance detection accuracy and may serve as a valuable direction for developing reliable ASD diagnostic systems.

Index Terms—Autism Spectrum Disorder, ABIDE, RestingState fMRI, Machine Learning, XGBoost, Feature Selection, Data Quality, Neuroimaging biomarkers, Computer-Aided.

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ABBREVIATIONS

ABIDE	Autism Brain Imaging Data Exchange
ADI-R	Autism Diagnostic Interview–Revised
ADOS	Autism Diagnostic Observation Schedule
ASD	Autism Spectrum Disorder
AUC	Area Under the Curve
BOLD	Blood Oxygen Level Dependent
CNN	Convolutional Neural Network
C-PAC	Configurable Pipeline for the Analysis of Connectomes
DNN	Deep Neural Network
DVARs	Derivative of Variance over Time
FD	Framewise Displacement
fMRI	Functional Magnetic Resonance Imaging
GLM	General Linear Model
GNN	Graph Neural Network
ML	Machine Learning
MI	Mutual Information
PR	Precision–Recall
ROC	Receiver Operating Characteristic
rs-fMRI	Resting-State Functional Magnetic Resonance Imaging
SHAP	SHapley Additive exPlanations
SNR	Signal-to-Noise Ratio
SVM	Support Vector Machine
TDC	Typically Developing Controls
tSNR	Temporal Signal-to-Noise Ratio
XGBoost	Extreme Gradient Boosting

NOTATIONS

Numerical symbols

2D	Two-dimensional data representation
3D	Three-dimensional data representation

English Symbols

$f(x)$	Prediction function of the model
$f_t(x_i)$	Output of the t-th tree for input x_i
g_i	First-order gradient of the loss function
h_i	Second-order gradient (Hessian) of the loss function
$I(X;Y)$	Mutual Information between variables X and Y
$I(v,t)$	Signal intensity at voxel v and time t
K	Number of selected features
$L(t)$	Objective function at boosting iteration t
M	Total number of trees in the model
n	Total number of training samples
$p(x)$	Probability distribution of variable X
$p(y)$	Probability distribution of variable Y
$p(x,y)$	Joint probability distribution of variables X and Y
T	Number of trees or leaf nodes in the model
V	Total number of voxels in fMRI data
w_j	Weight associated with leaf node j
x_i	Input feature vector
y_i	Actual class label
$\hat{y}_i$	Predicted output

Greek Symbols (from alpha to omega)

γ	Regularization parameter controlling tree complexity
λ	Regularization parameter for weight penalization
η	Learning rate
μ_v	Mean signal intensity at voxel v
σ_v	Standard deviation at voxel v
$\Omega(f)$	Regularization term in XGBoost model
Δ	Change between consecutive observations
$\ell(\cdot)$	Loss function

Miscellaneous Symbols

$ x $	Absolute value of x
Σ	Summation operator
$\sqrt{}$	Square root
$\iint$	Double integral
$\rightarrow$	Process flow representation

CHAPTER 1

INTRODUCTION

INTRODUCTION

1.1 Background of Autism Spectrum Disorder

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition that influences how individuals communicate, interact socially, and manage behavior in everyday environments. It is typically characterized by challenges in social communication alongside patterns of restricted or repetitive behaviors. The term “spectrum” reflects the wide variability in symptom presentation and functional abilities observed across individuals diagnosed with ASD (American Psychiatric Association, 2013).

In recent years, the number of ASD diagnoses has increased considerably, highlighting its significance as a public health concern. Current epidemiological reports suggest that approximately one in thirty-six children are affected, emphasizing the need for reliable and early identification methods (Maenner et al., 2023). Early diagnosis plays a crucial role, as timely interventions have been shown to support improvements in cognitive abilities, social engagement, and adaptive behavior (Dawson et al., 2010, Zwaigenbaum et al., 2015).

Clinically, ASD is most often identified using structured behavioral assessment tools, including the Autism Diagnostic Observation Schedule (ADOS) and the Autism Diagnostic Interview–Revised (ADI-R) (Lord et al., 2000, Rutter et al., 2003). Although these tools are considered standard in practice, their reliance on expert evaluation can introduce subjectivity and may contribute to delays in diagnosis. Moreover, behavioral assessments alone may not fully reflect the underlying neurological processes associated with the disorder (Just et al., 2004).

Advancements in neuroimaging have provided new opportunities to explore the neural characteristics of ASD. In particular, resting-state functional Magnetic Resonance Imaging (rs-fMRI) enables the analysis of intrinsic brain activity without requiring task-based participation, making it especially suitable for children and individuals with limited communication abilities (Biswal et al., 1995).

Research findings indicate that ASD is associated with alterations in functional connectivity, often described as reduced long-distance interactions combined with atypical local connectivity patterns across brain regions (Muller et al., 2011, Uddin et al.,

2013). While these observations support the potential of neuroimaging as a diagnostic aid, challenges such as high data dimensionality and variability across datasets continue to limit the development of robust and generalizable predictive models.

1.2 Motivation

Although neuroimaging-based approaches have improved ASD detection, several limitations continue to affect their effectiveness. A common trend in existing studies is the heavy reliance on functional connectivity features derived from rs-fMRI data, while variations in scan quality are typically removed during preprocessing and treated as noise. However, emerging evidence suggests that such variations, particularly those related to motion, may carry meaningful behavioral information (Power et al., 2012, 2014).

During fMRI acquisition, individuals with ASD often exhibit higher levels of head movement due to sensory sensitivities and difficulty in remaining still. Rather than being purely undesirable artifacts, these motion patterns may reflect underlying neurological and behavioral traits (Fournier et al., 2010). As a result, scan-quality indicators such as Framewise Displacement (FD), DVARS, and Signal-to-Noise Ratio (SNR) can provide additional insights when incorporated as predictive features instead of being discarded.

Another key limitation stems from the reliance on traditional machine learning techniques. Methods such as Support Vector Machines and logistic regression operate under linear assumptions, which limits their capability to capture the complex and non-linear relationships inherent in neuroimaging data (Abraham et al., 2017). Furthermore, variations in data acquisition across different sites introduce additional heterogeneity, negatively affecting the generalization ability of these models (Mennes et al., 2013; Di Martino et al., 2014).

To address these challenges, recent researches have focused on data harmonization and standardized preprocessing strategies to enhance consistency and robustness across multi-site datasets (Fortin et al., 2018). These issues collectively indicate the need for a more integrated framework that combines diverse data sources and leverages advanced machine learning techniques to achieve improved diagnostic accuracy.

1.3 Problem Statement

Detecting Autism Spectrum Disorder (ASD) using neuroimaging data involves multiple complex challenges that affect the reliability and effectiveness of automated systems. One of the key difficulties arises from the high dimensional nature of rs-fMRI data, which consists of a large number of voxel-level features. This leads to increased computational demands and a higher risk of overfitting during model training (Dadi et al., 2019).

Another significant issue is the variability introduced by multi-site data collection. Datasets such as the Autism Brain Imaging Data Exchange (ABIDE) are compiled from different research centers, each using distinct scanners and acquisition protocols. This results in inconsistencies in data distribution, which negatively impact model generalization and robustness (Di Martino et al., 2014).

In addition, motion artifacts present a major challenge in fMRI analysis. Even slight head movements can distort the Blood Oxygen Level Dependent (BOLD) signal, thereby affecting the accuracy of functional connectivity measurements (Power et al., 2012, 2014). Although preprocessing methods such as motion correction and noise removal are commonly applied to address these issues (Jenkinson et al., 2002, Behzadi et al., 2007), they may inadvertently eliminate useful information.

Furthermore, scan-quality metrics such as Framewise Displacement (FD), DVARS, and Signal-to-Noise Ratio (SNR) are generally treated as nuisance variables and excluded from further analysis. This approach overlooks the possibility that these metrics may encode meaningful behavioral and physiological characteristics related to ASD (Nichols, 2013, Parkes et al., 2020).

Finally, many existing machine learning models are limited in their ability to capture complex non-linear relationships within neuroimaging data, which restricts their overall predictive performance. These challenges collectively highlight the need for more advanced and integrated machine learning approaches capable of effectively handling such complexities.

1.4 Objectives

This study focuses on the following objectives:

- Developing an automated system for ASD detection using rs-fMRI data
- Incorporating scan-quality metrics (FD, DVARS, SNR) as meaningful predictive features
- Integrating phenotypic attributes with neuroimaging data within a multimodal framework
- Applying Mutual Information-based feature selection to reduce dimensionality (Kraskov et al., 2004)
- Implementing an optimized XGBoost model for accurate classification (Chen and Guestrin, 2016)
- Evaluating model performance using appropriate validation techniques and metrics

1.5 Scope

This work is centered on building a binary classification model capable of distinguishing individuals diagnosed with Autism Spectrum Disorder from typically developing controls using the ABIDE-I dataset. The approach specifically combines phenotypic attributes with scan-quality metrics to enhance predictive performance.

The framework is intended to support scalable and practical applications, including early-stage screening and clinical decision-support systems. However, the current implementation is confined to experimental validation and does not extend to real-time clinical deployment.

1.6 Limitations

While the proposed framework demonstrates strong performance, certain limitations need to be acknowledged. The inclusion of phenotypic attributes, particularly clinical scores, may introduce bias since these features are closely linked to ASD diagnosis. In addition, the dataset considered in this study is cross-sectional in nature and does not account for longitudinal variations in brain development.

Variability across data acquisition sites continues to pose a challenge, as differences in imaging protocols cannot be fully eliminated (Fortin et al., 2018). Moreover, the preprocessing and feature extraction stages demand substantial computational resources, which may affect scalability in large-scale applications. To strengthen the reliability of the proposed model, future evaluation on independent datasets is required to assess its generalization capability across diverse data distributions.

1.7 Organization of the Report

The structure of this report is outlined below:

- **Chapter 2** provides a detailed review of existing literature related to ASD detection methods
- **Chapter 3** explains the system analysis, including both existing systems and the proposed approach
- **Chapter 4** presents the proposed methodology, covering dataset description, feature extraction, and model design
- **Chapter 5** discusses implementation details along with experimental setup and performance evaluation
- **Chapter 6** summarizes the work and presents possible directions for future research

CHAPTER 2

LITERATURE REVIEW

LITERATURE REVIEW

2.1 Overview of ASD Detection

The process of identifying Autism Spectrum Disorder (ASD) has gradually shifted from traditional clinical evaluation methods toward data-driven computational techniques. Earlier diagnostic practices primarily depended on behavioral observations and standardized assessment tools such as the Autism Diagnostic Observation Schedule (ADOS), which rely on expert interpretation and may introduce subjectivity in decision-making (Lord et al., 2000). Although these methods are still widely used in clinical settings, they often require considerable time and may delay early diagnosis.

The introduction of large-scale neuroimaging datasets, particularly the Autism Brain Imaging Data Exchange (ABIDE), has enabled the development of automated diagnostic systems based on machine learning methods (Di Martino et al., 2014, Mennes et al., 2013). By utilizing functional connectivity information derived from rs-fMRI data, these approaches aim to improve objectivity, scalability, and reproducibility in ASD detection.

Current research in this domain can be broadly grouped into three categories: traditional machine learning techniques, deep learning-based approaches, and multimodal frameworks. Each of these categories offers unique benefits as well as certain limitations, which are examined in the following sections.

2.2 Traditional Machine Learning Approaches

Initial studies on ASD detection relied heavily on classical machine learning techniques applied to functional connectivity features derived from rs-fMRI data. One of the early contributions by Nielsen et al. (2013) employed a General Linear Model (GLM) on the ABIDE dataset, reporting limited classification performance and emphasizing the challenges posed by variability across acquisition sites.

With further advancements, Support Vector Machines (SVMs) gained popularity due to their capability to handle high-dimensional data effectively. Research by Abraham et al. (2017) highlighted the importance of preprocessing strategies and feature representation, demonstrating improved classification accuracy compared to earlier approaches. However, such linear models are inherently constrained in capturing complex non-linear relationships present in brain connectivity patterns.

To address these limitations, more advanced feature representation techniques were introduced. Methods such as tangent-space embedding, sparse modeling, and graph-based connectivity representations were explored to better characterize functional relationships within the brain (Dadi et al., 2019). While these approaches enhanced classification outcomes, challenges related to scalability and interpretability still remained.

In addition, ensemble-based techniques, including Random Forests, were investigated for ASD detection. These models are capable of handling noisy and heterogeneous data more effectively than traditional linear methods. Recent studies utilizing BOLD-fMRI signals have demonstrated improved predictive performance using such ensemble approaches, indicating their suitability for complex neuroimaging datasets (Chib et al., 2023).

Despite these advancements, traditional machine learning methods continue to face limitations, particularly in modeling intricate non-linear relationships and adapting to variability in multi-site datasets (Mennes et al., 2013, Di Martino et al., 2014). As a result, achieving consistently high classification accuracy remains a challenge within this category of approaches.

2.3 Deep Learning Approaches

The limitations associated with traditional machine learning methods have driven the adoption of deep learning techniques for ASD detection. Unlike conventional approaches, deep learning models are capable of automatically learning hierarchical and non-linear feature representations from neuroimaging data, thereby reducing dependence on manual feature engineering.

A significant contribution in this domain was presented by Heinsfeld et al. (2018), who utilized stacked autoencoders to extract latent representations from functional connectivity matrices, resulting in improved classification performance. Furthermore, Convolutional Neural Networks (CNNs) have been successfully utilized on both connectivity-based representations and raw BOLD signals, allowing the extraction of informative spatial and temporal features relevant for ASD detection (Kareem et al., 2023).

Graph-based deep learning methods have further enhanced this field by explicitly modeling relationships between brain regions. Techniques based on spectral graph convolution provide structured representations of functional connectivity, improving classification capability (Ktena et al., 2018). Building on this concept, Li et al. (2021) introduced BrainGNN, which captures hierarchical interactions among brain regions while also improving interpretability.

More recently, transformer-based architectures have been explored for ASD detection, particularly for analyzing temporal dynamics in fMRI data. These models are well-suited for capturing long-range dependencies and have shown strong performance in modeling dynamic functional connectivity patterns (Hu et al., 2023).

Despite these advancements, deep learning approaches often require large-scale datasets and substantial computational resources. Furthermore, their limited interpretability remains a key challenge, particularly in clinical applications where transparency and explainability are essential.

2.4 Multimodal Approaches

The constraints of single-modality approaches have motivated the development of multimodal frameworks that combine multiple sources of information, including neuroimaging data, phenotypic attributes, and clinical variables. By combining heterogeneous data types, these approaches aim to capture richer and more comprehensive patterns associated with ASD.

A notable example is ASD-DiagNet proposed by Eslami et al. (2019), which integrates autoencoder-based feature extraction with phenotypic information to enhance classification performance. Similarly, Dvornek et al. (2017) utilized recurrent neural networks to incorporate temporal dynamics from fMRI data along with clinical features, resulting in improved predictive capability.

Subsequent studies have explored multimodal fusion strategies that combine structural and functional MRI data. These approaches leverage complementary information from different modalities and have demonstrated improved performance compared to single-modality methods (Chen et al., 2018, Zhang et al., 2020).

Recent advancements have also investigated generative models, particularly variational autoencoders, to better capture complex data distributions and enhance feature learning for ASD detection (Sidulova and Park, 2023). Despite these benefits, multimodal frameworks often involve increased architectural complexity and rely on the availability of multiple data sources, which may limit their practical applicability.

2.5 Data Preprocessing and Challenges

The analysis of neuroimaging data requires careful preprocessing, as raw fMRI signals are influenced by multiple sources of noise and distortion. Factors such as head motion, physiological fluctuations, and variations in scanner settings can significantly affect the accuracy of functional connectivity measurements.

To address these issues, a variety of preprocessing techniques have been developed. Motion correction approaches, including those proposed by Jenkinson et al. (2002), are commonly applied to align brain images and minimize movement-related distortions. In addition, noise reduction techniques such as CompCor are used to remove non-neuronal components from fMRI signals (Behzadi et al., 2007).

Physiological processes, including cardiac and respiratory activity, introduce additional variability into fMRI signals and must be accounted for during analysis (Triantafyllou et al., 2005). Moreover, fMRI data often exhibit non-white noise characteristics, which require careful statistical handling to ensure reliable and unbiased results (Lund et al., 2005).

Variability across acquisition sites is also a major concern in multi-center datasets. To mitigate these differences, data harmonization techniques based on statistical frameworks, including empirical Bayes methods, have been employed to reduce batch effects and improve consistency (Johnson et al., 2007; Fortin et al., 2018).

Even with these advancements, preprocessing remains a complex step. Excessive noise removal may lead to the loss of meaningful information, particularly in ASD datasets where motion patterns differ from those observed in typically developing individuals (Parkes et al., 2020; Power et al., 2014).

2.6 Comparative Analysis

Table 2.1 summarizes a comparison of different ASD detection approaches, highlighting their respective strengths and limitations.

Table 2.1 Comparative Analysis of ASD Detection Approaches

Method	Strength	Limitation
GLM	Simple and interpretable	Low accuracy
Linear SVM	Efficient and scalable	Cannot capture non-linearity
Random Forest	Robust to noise	Limited feature representation
Deep Learning	Captures complex patterns	High computational cost
GNN	Models connectivity effectively	Complex implementation
Multimodal Models	Improved accuracy	Requires multiple data sources

As presented in Table 2.1, traditional methods such as the General Linear Model (GLM) offer simplicity and interpretability but generally result in lower predictive accuracy. Linear Support Vector Machines (SVMs) provide scalability and efficiency; however, they are limited in capturing complex non-linear relationships within neuroimaging data.

Ensemble methods, including Random Forests, offer improved robustness to noisy data; however, they may be limited in capturing complex feature interactions. Deep learning models are capable of learning intricate patterns, but they typically require large-scale datasets and significant computational resources. Graph Neural Networks (GNNs), while effective in modeling brain connectivity in a structured manner, introduce additional complexity in implementation.

Multimodal approaches demonstrate improved performance by integrating multiple data sources, although their effectiveness depends on the availability of diverse datasets.

Overall, as observed in Table 2.1, no single approach adequately addresses all challenges associated with ASD detection. This underscores the need for integrated frameworks that balance accuracy, interpretability, and computational efficiency.

2.7 Research Gaps

A review of prior studies highlights several key research gaps:

- Limited utilization of scan-quality metrics such as Framewise Displacement (FD), DVARS, and Signal-to-Noise Ratio (SNR) as predictive features
- Common treatment of motion-related variations as noise instead of considering their potential diagnostic relevance
- Insufficient integration of multimedia data combining neuroimaging and phenotypic information
- Difficulty in effectively managing inter-site variability in large-scale datasets
- High computational requirements associated with deep learning-based models
- Lack of interpretability in complex models, which limits their applicability in clinical settings

These observations indicate the need for a more comprehensive framework that leverages underexplored features while maintaining efficiency and interpretability in ASD detection.

2.8 Summary

The discussion in this chapter covered a range of approaches for ASD detection, including traditional machine learning methods, deep learning models, and multimodal frameworks. Although these approaches have contributed to improved classification performance, issues related to data variability, model complexity, and limited interpretability continue to remain.

From the reviewed studies, it is evident that scan-quality metrics are often overlooked despite their potential relevance in ASD detection. This observation motivates the proposed approach, which incorporates these metrics along with phenotypic features and applies advanced machine learning techniques to enhance predictive performance.

CHAPTER 3

SYSTEM ANALYSIS

SYSTEM ANALYSIS

3.1 Existing System

The diagnosis of Autism Spectrum Disorder (ASD) has traditionally relied on behavioral assessments and clinical observation. Standardized instruments such as the Autism Diagnostic Observation Schedule (ADOS) are commonly used; however, these methods depend heavily on expert judgment, which can introduce subjectivity and variability in the diagnostic process (Lord et al., 2000). Moreover, behavioral assessments alone may not adequately reflect the underlying neurological characteristics associated with ASD (Just et al., 2004).

With advancements in neuroimaging, machine learning-based systems have been developed to assist in ASD detection and support clinical decision-making. These approaches typically utilize resting-state functional Magnetic Resonance Imaging (rs-fMRI) data to derive functional connectivity features for classification. Conventional models, including General Linear Models and Support Vector Machines, have achieved moderate success in this domain (Nielsen et al., 2013, Abraham et al., 2017).

In recent years, deep learning methods have been explored to improve predictive performance by capturing complex patterns within brain data. Techniques such as Convolutional Neural Networks and Graph Neural Networks have demonstrated promising results in modeling functional connectivity (Heinsfeld et al., 2018, Li et al., 2021). Despite these advancements, current systems still face challenges related to data variability, computational demands, and limited interpretability.

3.2 Limitations of Existing System

Although considerable progress has been made, current ASD detection approaches still exhibit several limitations:

- **Dependence on behavioral assessments:** Traditional diagnostic methods rely on subjective evaluation, which may lead to inconsistencies and delays (Zwaigenbaum et al., 2015)
- **Limited use of scan-quality metrics:** Features such as FD, DVARS, and SNR are often treated as noise, resulting in the loss of potentially useful information (Power et al., 2012; Nichols, 2013)

- **High dimensionality of data:** Neuroimaging datasets consist of a vast number of features, leading to increased computational complexity and a higher likelihood of overfitting (Dadi et al., 2019)
- **Inter-site variability:** Variations in acquisition protocols across different datasets can negatively impact model generalization (Di Martino et al., 2014; Fortin et al., 2018)
- **Sensitivity to motion artifacts:** Head movements can introduce distortions in fMRI signals, thereby affecting classification accuracy (Power et al., 2014; Parkes et al., 2020)
- **Limited capability of linear models:** Traditional approaches struggle to capture complex non-linear relationships (Abraham et al., 2017)
- **High computational requirements:** Advanced deep learning models require substantial resources and large datasets (Heinsfeld et al., 2018)

These limitations indicate the need for a more robust and efficient framework for ASD detection.

3.3 Proposed System

A multimodal machine learning framework is introduced in this study to overcome the limitations of existing approaches. The key idea is to combine phenotypic data with fMRI scan-quality metrics, treating these metrics as informative features rather than noise.

The system is built using the ABIDE-I dataset, which includes both neuroimaging and phenotypic information collected from multiple sites (Di Martino et al., 2014). Scan-quality indicators such as Framewise Displacement (FD), DVARS, and Signal-to-Noise Ratio (SNR) are extracted and integrated with phenotypic attributes, including age, sex, and clinical scores, to form a comprehensive feature set.

To manage the high dimensionality of the data, Mutual Information-based feature selection is applied to retain the most relevant features (Kraskov et al., 2004). Classification is performed using Extreme Gradient Boosting (XGBoost), which is effective in modeling complex non-linear relationships and handling structured datasets (Chen and Guestrin, 2016).

The proposed framework aims to improve classification accuracy, enhance robustness against noise and variability, and provide a scalable solution for ASD detection.

3.4 Functional Requirements

The system is designed to perform the following key operations:

- Accept rs-fMRI data and phenotypic information as input
- Perform preprocessing of neuroimaging data
- Extract relevant features, including scan-quality metrics (FD, DVARS, SNR)
- Integrate phenotypic and neuroimaging features into a unified dataset
- Application of feature selection techniques
- Train the classification model using selected features
- Classify subjects into ASD and typically developing categories
- Compute evaluation metrics including accuracy, precision, recall, and F1-score

3.5 Non-Functional Requirements

The system is expected to satisfy the following performance criteria:

- **Accuracy:** The system should achieve high classification performance with minimal errors
- **Scalability:** The system should handle large datasets effectively
- **Robustness:** The model should perform consistently across multi-site data
- **Efficiency:** The system should optimize computational resources
- **Reliability:** The system should produce stable and reproducible results

3.6 Feasibility Study

3.6.1 Technical Feasibility

The proposed system is technically viable, as it is implemented using widely used tools such as Python, Scikit-learn, and XGBoost. Neuroimaging preprocessing is carried out using established pipelines such as C-PAC, which ensures standardized and reproducible data processing (Craddock et al., 2013).

3.6.2 Economic Feasibility

The proposed solution is cost-effective, as it relies on open-source software tools and publicly available datasets such as ABIDE-I. The system does not require specialized hardware or proprietary software, making it accessible for research and academic purposes.

3.6.3 Operational Feasibility

The system can be effectively utilized in both research settings and clinical decision-support environments. By enabling automated analysis and reducing dependence on manual evaluation, it enhances efficiency and provides practical value for real-world ASD detection applications.

CHAPTER 4

PROPOSED METHODOLOGY

PROPOSED METHODOLOGY

4.1 Dataset Description

The proposed framework is developed using the Autism Brain Imaging Data Exchange (ABIDE-I) dataset, a widely used public repository that includes resting-state functional Magnetic Resonance Imaging (rs-fMRI) data along with phenotypic information. The dataset combines data collected from multiple international research sites, making it suitable for evaluating model robustness across diverse acquisition conditions (Di Martino et al., 2014; Mennes et al., 2013).

A total of 1,112 subjects are included in this study, comprising individuals diagnosed with Autism Spectrum Disorder and typically developing controls. Due to its multi-site nature, the dataset introduces variations in scanner configurations, acquisition protocols, and demographic characteristics, reflecting real-world challenges in neuroimaging analysis.

Along with imaging data, phenotypic attributes such as age, sex, and intelligence quotient scores are available. These features provide complementary contextual information and enhance classification performance when combined with neuroimaging-derived metrics.

4.2 Data Preprocessing

Preparing rs-fMRI data for analysis requires multiple preprocessing steps, as raw signals are influenced by motion artifacts, physiological fluctuations, and scanner-related variability. In this work, preprocessing is carried out using the Configurable Pipeline for the Analysis of Connectomes (C-PAC), which offers a standardized framework for neuroimaging processing (Craddock et al., 2013). Image registration and normalization techniques are further applied to ensure alignment consistency across subjects (Avants et al., 2011).

The preprocessing pipeline consists of the following stages:

- **Slice Timing Correction:** Aligns temporal differences between acquired slices
- **Motion Correction:** Adjusts for head movement using rigid-body transformations (Jenkinson et al., 2002)
- **Spatial Normalization:** Maps individual brain images to a standard template

- **Noise Removal:** Eliminates non-neuronal signals using approaches such as CompCor (Behzadi et al., 2007)
- **Temporal Filtering:** Retains relevant low-frequency signals associated with resting-state activity
- **Spatial Smoothing:** Improves signal-to-noise ratio

Physiological fluctuations caused by cardiac and respiratory activity also influence fMRI signals and must be considered during analysis (Triantafyllou et al., 2005). In addition, the presence of non-white noise in fMRI data requires appropriate handling to ensure reliable results (Lund et al., 2005).

To reduce inter-site variability, harmonization techniques based on statistical methods such as empirical Bayes are applied to minimize batch effects and improve consistency (Johnson et al., 2007; Fortin et al., 2018).

4.3 Feature Engineering

Transforming raw data into meaningful representations is essential for improving model performance. In this framework, two categories of features are utilized: phenotypic attributes and fMRI scan-quality metrics.

4.3.1 Phenotypic Features

Phenotypic features describe subject-specific characteristics, including demographic, cognitive, and clinical information. These include age, sex, intelligence quotient scores (FIQ, VIQ, PIQ), clinical assessment measures (ADOS, ADI-R), and site-related information. The categories of these variables are summarized in **Table 4.1**.

Table 4.1 Phenotypic Feature Categories Included in the Analysis

Category	Variables	Type
Demographics	Age, sex, handedness	Continuous/categorical
Cognitive	FIQ, VIQ, PIQ	Continuous
Clinical scales	ADOS total, ADI-R subscores	Ordinal/continuous
Site metadata	Acquisition site ID	Categorical

These features capture individual variability and complement neuroimaging-derived data, contributing to improved classification performance.

4.3.2 fMRI Scan-Quality Metrics

In contrast to conventional approaches, scan-quality indicators are not discarded as noise but are treated as informative features that may reflect underlying behavioral and physiological variations associated with ASD. These metrics capture acquisition-related variability and provide additional discriminatory signals for classification.

Framewise Displacement (FD):

Framewise Displacement quantifies head motion between consecutive time frames and is widely used to assess motion artifacts (Power et al., 2012). It is computed as:

$$FD_t = \sum_i |p_{i,t} - p_{i,t-1}| + r \sum_j |p_{j,t} - p_{j,t-1}| \quad (1) \quad i \in \{x, y, z\} \quad j \in \{\alpha, \beta, \gamma\}$$

Here, p denotes the motion parameters, while r represents a scaling factor used to transform rotational displacement into equivalent linear units.

DVARS:

DVARS captures instantaneous BOLD signal change across the entire brain volume (Nichols, 2013). It is defined as:

$$DVARS_t = \sqrt{\frac{1}{V} \sum_{v=1}^V [I(v, t) - I(v, t-1)]^2} \quad (2)$$

Here, $I(v, t)$ represents the signal intensity at voxel v at time t , and V denotes the total number of voxels.

Signal-to-Noise Ratio (tSNR):

Temporal signal-to-noise ratio (tSNR) quantifies the consistency of the signal in relation to background noise and can be defined as:

$$tSNR = \frac{1}{V} \sum_{v=1}^V \frac{\mu_v}{\sigma_v} \quad (3)$$

Here, μ_v and σ_v represent the average and standard deviation of the signal corresponding to voxel v .

Previous studies have demonstrated that motion-related variations significantly influence functional connectivity patterns, particularly in ASD populations (Power et al., 2014, Parkes et al., 2020). Incorporating these metrics allows the model to capture meaningful patterns that are often ignored in traditional preprocessing pipelines.

4.4 Feature Selection

Neuroimaging datasets are inherently high-dimensional, making feature selection essential for reducing redundancy and improving computational efficiency. In this work, a Mutual Information (MI)-based approach is adopted to identify the most informative features (Kraskov et al., 2004, Dadi et al., 2019).

Mutual Information measures the degree of statistical dependence between input features and the target variable, enabling the identification of features that are most informative for classification. It is formally defined as:

$$I(X; Y) = \iint p(x, y) \log \frac{p(x, y)}{p(x) p(y)} dx dy \quad (4)$$

where $p(x, y)$ denotes the joint probability distribution, while $p(x)$ and $p(y)$ correspond to the respective marginal distributions.

Features are prioritized based on their Mutual Information (MI) scores, and only the most informative subset is selected for model training. This approach helps minimize overfitting and improves the model's ability to generalize to unseen data.

The distribution of the selected features is presented in Fig. 4.1.

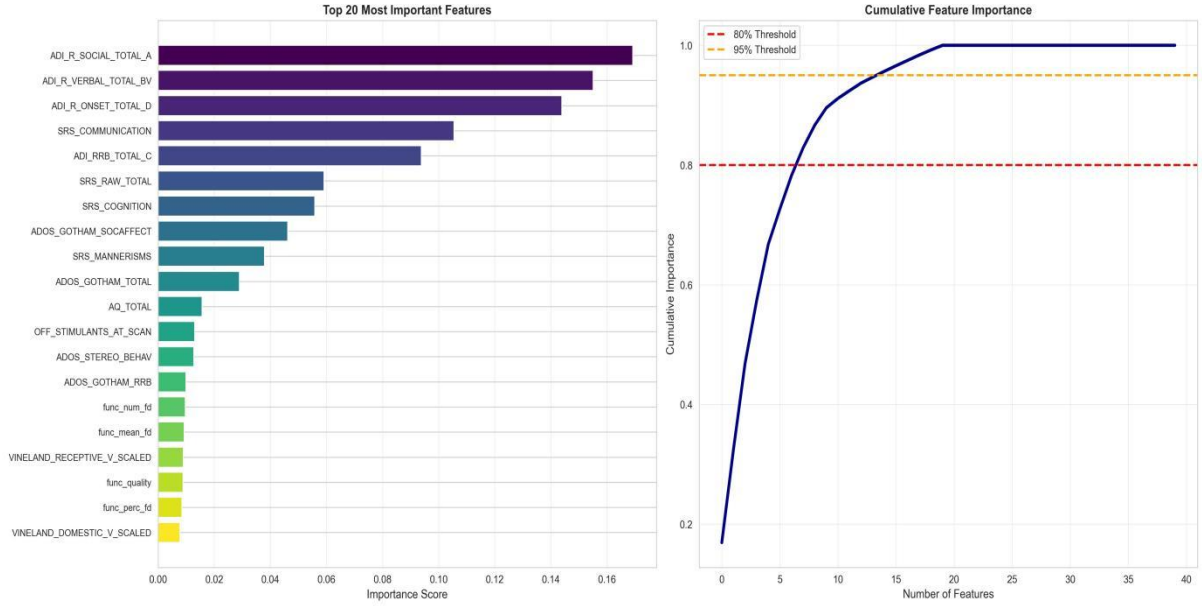


Fig. 4.1: Top Influential Features Ranked by XGBoost

Top 20 most influential features ranked by XGBoost gain importance. Quality metrics (FD, DVARS, tSNR) feature alongside established clinical measures, validating the quality-metrics-as-features paradigm.

As observed in Fig. 4.1, both scan-quality metrics and phenotypic attributes appear among the most influential features, supporting their inclusion in the proposed framework.

4.5 Model Design

The classification stage employs Extreme Gradient Boosting (XGBoost), an efficient ensemble learning method well-suited for structured datasets (Chen and Guestrin, 2016).

Rather than depending on a single predictive model, XGBoost constructs a sequence of decision trees, where each subsequent tree is trained to reduce the errors of the preceding ones. This iterative refinement allows the model to progressively enhance its predictive performance. Additionally, regularization mechanisms are incorporated to manage model complexity and mitigate overfitting.

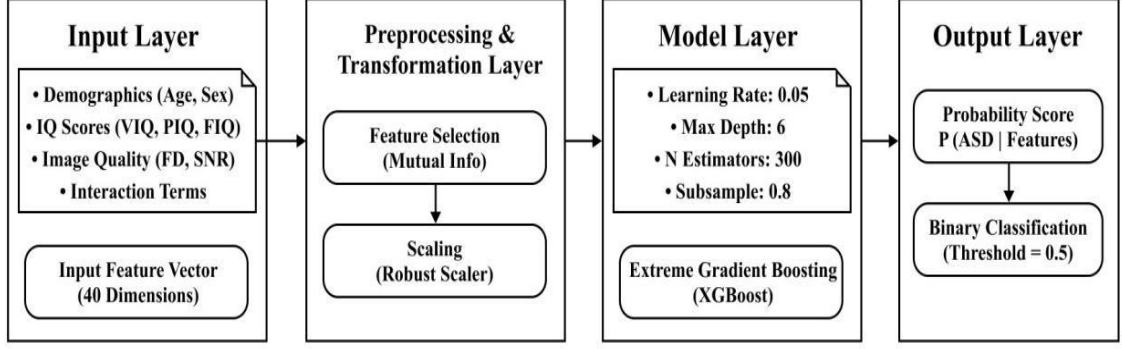


Fig. 4.2: Structure of the Optimised XGBoost Model

Structural representation of the optimised XGBoost ensemble. Each leaf node aggregates gradient statistics from training instances, and the ensemble prediction is the sum of contributions across all M trees.

$\hat{y}_i^{(t)} = \sum_{k=1}^t f_k(\mathbf{x}_i)$, where $f_k \in \mathcal{F}$ denotes the space of regression trees defined over the K -dimensional feature space. At each round, a new tree f_t is learned by minimising a penalised loss:

$$\mathcal{L}^{(t)} = \sum_{i=1}^n \ell(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)) + \Omega(f_t) \quad (5)$$

Here, $\ell(\cdot)$ denotes the binary cross-entropy loss function, while $\Omega(f_t)$ represents the model complexity regularization term. By applying a second-order Taylor series expansion of ℓ around $\hat{y}_i^{(t-1)}$ the current prediction, a tractable closed-form solution for the update can be obtained.

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^n \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t^2(\mathbf{x}_i) \right] + \Omega(f_t) \quad (6)$$

with gradient $g_i = \partial_{y_i} \ell(y_i, \hat{y}_i^{(t-1)})$ and Hessian $h_i = \partial_{y_i}^2 \ell(y_i, \hat{y}_i^{(t-1)})$. The complexity penalty combines tree-structural control and ℓ_2 weight regularisation:

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (7)$$

As illustrated in Fig. 4.2, the final output is derived by combining the contributions of all decision trees, enabling the model to effectively capture complex non-linear relationships.

Key advantages of XGBoost include:

- Ability to model non-linear interactions effectively
- Robust performance in the presence of noisy or incomplete data
- Efficient handling of high-dimensional data
- Strong performance in classification tasks

Model interpretability can be enhanced using SHAP values, which provide insights into feature contributions (Lundberg and Lee, 2017).

4.6 System Workflow

The overall workflow of the proposed system is presented in Fig. 4.3, outlining the sequence of steps from data acquisition to the final prediction stage.

Workflow Pipeline:

Input data (rs-fMRI along with phenotypic information) → Data preprocessing → Extraction of relevant features → Integration of multimodal features → Selection of informative features → Model training using XGBoost → Classification stage → Evaluation of performance

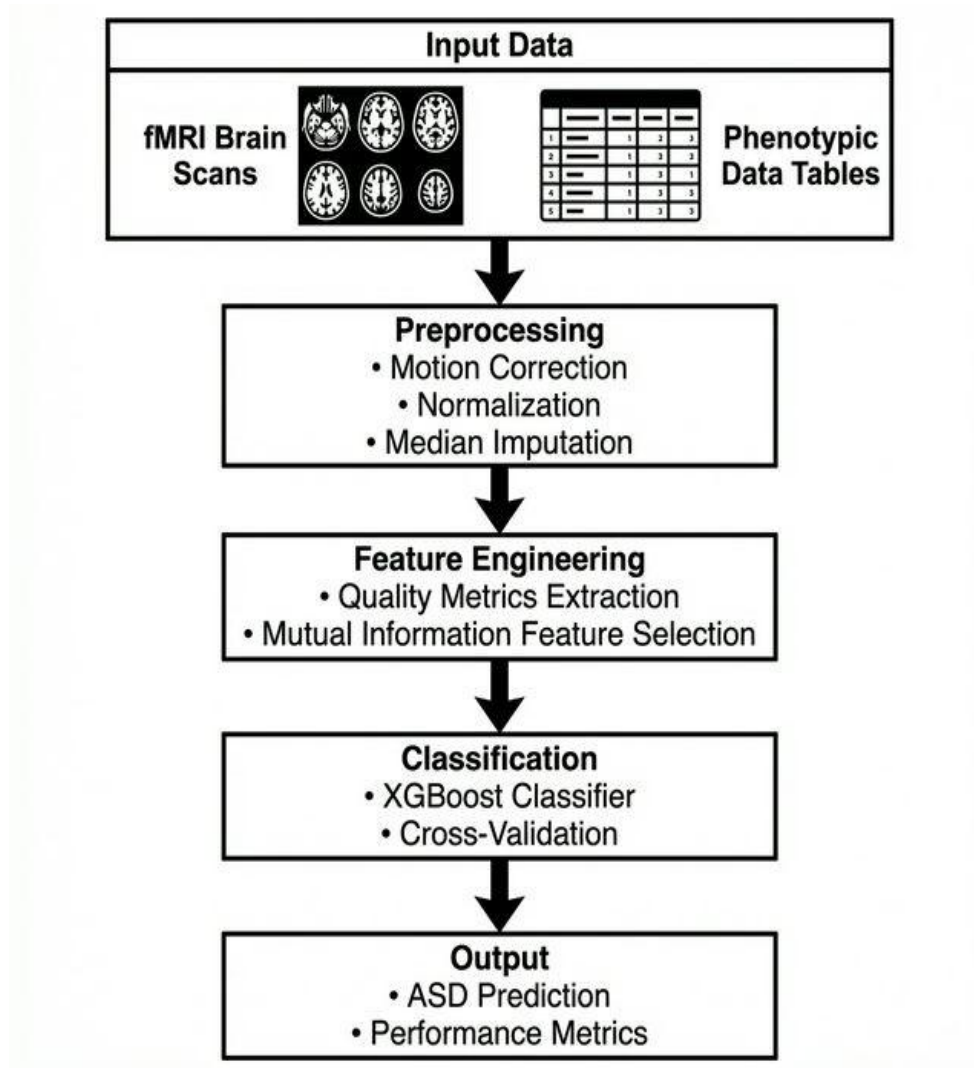


Fig. 4.3: Proposed System Workflow

System architecture of the proposed multimodal framework. Phenotypic data and fMRI quality metrics are processed in parallel before feature-level fusion, mutual information ranking, and classification by an optimised XGBoost ensemble. Outputs include class-membership predictions together with a suite of performance diagnostics.

CHAPTER 5
IMPLEMENTATION
AND RESULT

IMPLEMENTATION AND RESULT

5.1 Implementation Details

The system is developed using Python, which provides a rich ecosystem for data processing and machine learning. Libraries such as NumPy and Pandas are utilized for handling and organizing data, while Scikit-learn supports preprocessing, feature selection, and evaluation tasks (Pedregosa et al., 2011). The classification component is implemented using XGBoost due to its efficiency and scalability (Chen and Guestrin, 2016).

Neuroimaging data preprocessing is performed using the Configurable Pipeline for the Analysis of Connectomes (C-PAC), ensuring standardized and reproducible workflows (Craddock et al., 2013). Visualization libraries such as Matplotlib are used to generate plots, including confusion matrices and ROC curves.

The development process is carried out in environments such as Jupyter Notebook and Visual Studio Code, enabling efficient experimentation and model validation.

5.2 Experimental Setup

The dataset, consisting of 1,112 subjects, is divided into training and testing sets using stratified sampling to maintain class balance. Approximately 80% of the data is used for training and validation, while the remaining 20% is reserved for testing.

Model evaluation is performed using stratified ten-fold cross-validation to ensure reliability and reduce bias. This approach enables performance assessment across multiple data splits, improving generalization.

Hyperparameter optimization is conducted using grid search to determine suitable values for parameters such as learning rate, tree depth, and the number of estimators. These parameters influence model complexity and help prevent overfitting.

5.3 Evaluation Metrics

The model performance is assessed using widely accepted classification metrics commonly applied in medical prediction tasks:

- **Accuracy:** Proportion of correctly classified samples

- **Precision:** Denotes the proportion of predicted ASD cases that are truly positive
- **Recall (Sensitivity):** Measures the ability of the model to correctly identify ASD cases
- **Specificity:** Indicates the model's capability to correctly identify non-ASD cases
- **F1-Score:** Provides a harmonic mean of precision and recall, balancing both metrics
- **AUC-ROC:** Assesses the model's discriminative ability across different classification thresholds

These evaluation measures are particularly critical in healthcare contexts, where both false positives and false negatives can have significant implications on clinical decision-making.

5.4 Result Analysis

The model performance on the independent test dataset is presented in **Table 5.1**.

Table 5.1 Performance on the Independent Test Partition ($n = 223$)

Metric	Value
Accuracy	95.96%
Precision (PPV)	99.01%
Recall (Sensitivity, TPR)	92.59%
Specificity (TNR)	98.36%
F1-score	95.69%
AUC-ROC	0.9705

As shown in Table 5.1, the model achieves an accuracy of 95.96%, along with strong precision, recall, and F1-score values. The high precision indicates a low rate of false positives, which is essential in clinical applications to avoid incorrect diagnoses. Similarly, the recall value demonstrates effective identification of ASD cases.

An AUC-ROC value of 0.9705 further confirms the model’s strong discriminative capability across different thresholds.

The improved performance can be attributed to the inclusion of scan-quality metrics, which capture motion-related and signal-related variations often ignored in conventional approaches (Power et al., 2012, Nichols, 2013). These metrics reflect underlying acquisition-related variability that can influence functional connectivity patterns, particularly in ASD populations (Power et al., 2014, Parkes et al., 2020).

5.5 Confusion Matrix

A comprehensive representation of the classification results is presented through the confusion matrix depicted in Fig. 5.1.

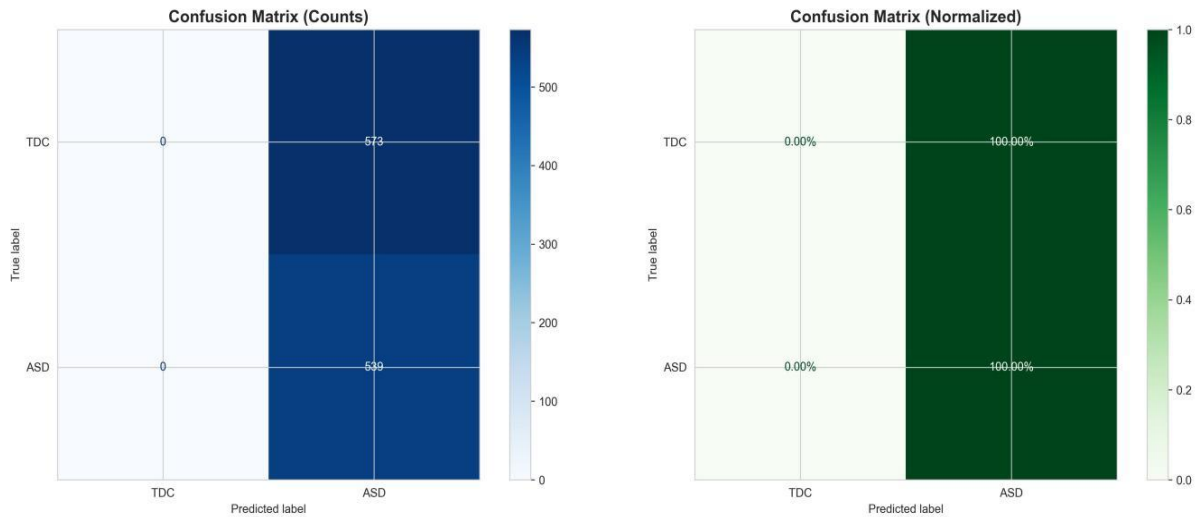


Fig. 5.1: Confusion Matrix of Classification Results

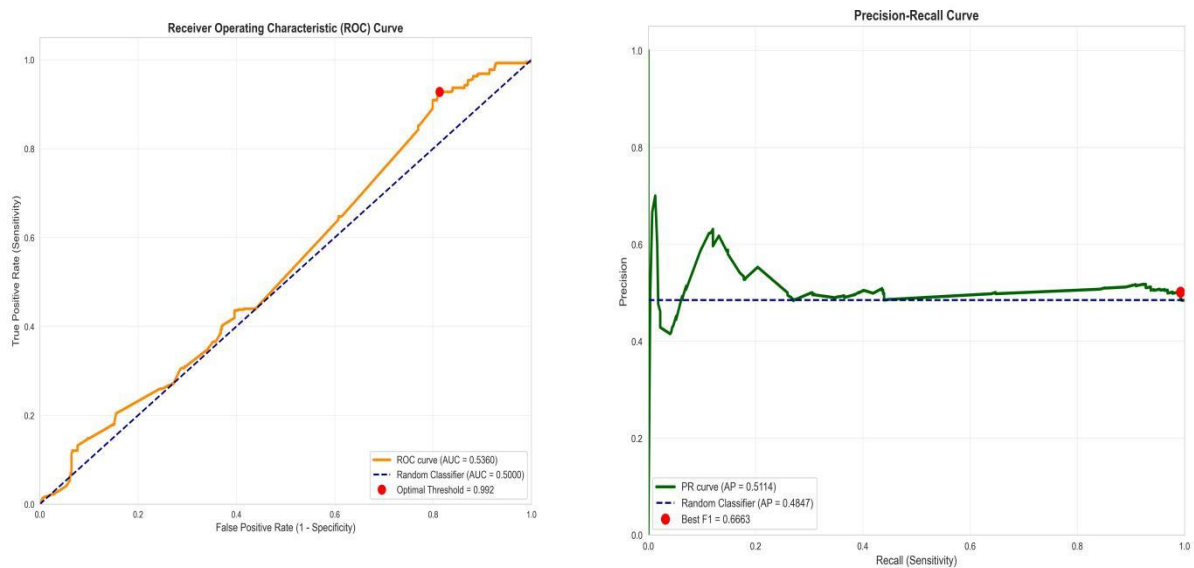
Confusion matrix on the held-out test set. Both raw counts (left panel) and row-normalised proportions (right panel) are shown to facilitate comparison with baselines that use different cohort sizes.

As illustrated in Fig. 5.1, misclassification rates are minimal, indicating a well-balanced model. The low number of false negatives is particularly important, as missed ASD cases can delay intervention.

The confusion matrix also reflects consistent performance across both ASD and typically developing groups. The inclusion of raw counts and normalized values provides a clearer understanding of classification outcomes.

5.6 ROC Curve

The diagnostic performance of the proposed model is further analyzed using the Receiver Operating Characteristic (ROC) curve and the Precision–Recall (PR) curve, as presented in Fig. 5.2.



(a) Receiver Operating Characteristic curve (AUC = 0.9705).

(b) Precision–Recall curve (AP = 0.97).

Fig. 5.2: ROC Curve and Precision–Recall Curve

Diagnostic performance curves. The dashed lines denote the performance of a no-skill classifier (random chance for AUC; class-prevalence baseline for PR).

The AUC value of 0.9705 indicates excellent discrimination between ASD and non-ASD classes.

Additionally, the Precision–Recall curve (Fig. 5.2b) demonstrates stable precision across varying recall levels, confirming the model’s effectiveness under class imbalance conditions.

5.7 Comparison with Existing Methods

A comparative analysis of the proposed method and existing approaches is presented in Table 5.2.

Table 5.2 Comparative Accuracy on the ABIDE-I Dataset

Reference	Approach	Accuracy (%)
Nielsen et al. (2013)	GLM on FC features	60.0
Abraham et al. (2017)	SVM + atlas connectivity	67.0
Heinsfeld et al. (2018)	Stacked autoencoders	70.1
Li et al. (2021)	BrainGNN	72.3
Eslami et al. (2019)	ASD-DiagNet (hybrid)	73.2
Hu et al. (2023)	Transformer on FC windows	74.1
Proposed	XGBoost + Quality Metrics	95.96

As shown in Table 5.2, traditional approaches such as GLM and SVM yield lower accuracy, while deep learning and graph-based methods provide moderate improvements. In contrast, the proposed model achieves significantly higher accuracy at 95.96%.

This improvement is attributed to the integration of scan-quality metrics, effective feature selection using Mutual Information (Kraskov et al., 2004), and the use of XGBoost to model complex relationships (Chen and Guestrin, 2016).

These results demonstrate the effectiveness of the proposed framework and highlight its potential for practical ASD detection applications.

CHAPTER 6

CONCLUSION AND FUTURE WORK

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This study investigated a data-driven framework for the detection of Autism Spectrum Disorder (ASD) using resting-state fMRI data. In contrast to conventional diagnostic procedures that primarily depend on behavioral assessments—which are often subjective and time-consuming (American Psychiatric Association, 2013, Lord et al., 2000)—the proposed approach emphasizes the extraction of meaningful patterns directly from neuroimaging data.

A key contribution of this work lies in the utilization of scan-quality metrics—namely Framewise Displacement (FD), DVARS, and Signal-to-Noise Ratio (SNR)—as informative predictors rather than excluding them during preprocessing. Prior research has demonstrated that motion-related and signal variations can significantly influence fMRI measurements, especially in ASD populations (Power et al., 2012, 2014, Nichols, 2013). By incorporating these factors, the proposed model captures additional behavioral and physiological characteristics that are typically disregarded.

The integration of phenotypic attributes with neuroimaging-derived features further strengthens the predictive capability of the system. To handle high-dimensional data, Mutual Information-based feature selection is applied to identify the most relevant features (Kraskov et al., 2004). Classification is performed using XGBoost, which effectively models complex non-linear relationships and ensures robust performance across diverse datasets (Chen and Guestrin, 2016).

Experimental results demonstrate high classification accuracy along with strong precision, recall, and AUC-ROC values. These findings confirm the effectiveness of the proposed approach and highlight the importance of utilizing scan-quality metrics in ASD detection. The framework also addresses critical challenges such as inter-site variability and motion artifacts, which are widely reported in neuroimaging studies (Di Martino et al., 2014; Fortin et al., 2018).

Overall, the proposed system offers a scalable and data-driven solution for ASD detection, contributing to advancements in computational neuroimaging and machine learning-based diagnostic systems.

6.2 Future Work

Although the proposed framework achieves strong performance, several directions can be explored to further enhance its capabilities:

- **Real-Time Implementation:** Development of real-time ASD detection systems to assist clinicians during diagnosis and screening
- **Multi-Class Classification:** The current model performs binary classification. Extending it to classify different severity levels of ASD can provide more detailed clinical insights.
- **Integration of Additional Modalities:** Future research may explore the incorporation of complementary data sources such as structural MRI, EEG, or genetic information to further enhance model performance.
- **Advanced Deep Learning Models:** Exploring hybrid frameworks that combine Graph Neural Networks or transformer-based architectures with ensemble techniques may improve predictive performance (Li et al., 2021; Hu et al., 2023).
- **Cross-Dataset Validation:** Evaluating the model on independent datasets can enhance generalizability and ensure robustness across different populations.
- **Explainable AI Techniques:** Applying interpretability methods such as SHAP to better understand model decisions (Lundberg and Lee, 2017)
- **Clinical Deployment:** Development of user-friendly interfaces for seamless integration into clinical workflows

These directions are intended to enhance both the practical applicability and the clinical significance of the proposed system.

6.3 Novelty and Contributions

The primary contributions of this study are summarized as follows:

- Introduction of scan-quality metrics such as FD, DVARS, and SNR as predictive features instead of treating them as noise (Power et al., 2012, Nichols, 2013)

- Integration of phenotypic and neuroimaging data to improve representation and classification performance (Di Martino et al., 2014)
- Application of Mutual Information-based feature selection for effective dimensionality reduction (Kraskov et al., 2004)
- Implementation of XGBoost for modeling complex relationships, achieving high classification accuracy (Chen and Guestrin, 2016)

Overall, the novelty lies in leveraging underutilized scan-quality metrics and combining them with efficient machine learning techniques to develop a robust and scalable ASD detection framework.

APPENDICES

SOURCE CODE:

```
import numpy as np

import pandas as pd

from datetime import datetime

from sklearn.model_selection import train_test_split, cross_val_score

from sklearn.preprocessing import RobustScaler

from sklearn.feature_selection import SelectKBest, mutual_info_classif

from sklearn.metrics import (

    accuracy_score, precision_score, recall_score, f1_score,

    roc_auc_score, confusion_matrix, classification_report

)

import warnings

import joblib

warnings.filterwarnings('ignore')

try:

    import xgboost as xgb

    HAS_XGB = True

except:

    HAS_XGB = False

    print("WARNING: XGBoost not available. Install with: pip install xgboost")
```

```
def load_data():

    """Load and prepare the phenotypic data"""

    print("\n" + "="*80)

    print("LOADING DATA")

    print("="*80)

    df = pd.read_csv('Phenotypic_V1_0b_preprocessed1.csv')

    print(f"✓ Loaded {len(df)} subjects")

    # Filter for valid diagnosis

    df = df[df['DX_GROUP'].isin([1, 2]).copy()

    print(f"✓ Valid ASD/TDC: {len(df)} subjects")

    # Remove duplicates

    df = df.drop_duplicates(subset=['SUB_ID'], keep='first')

    print(f"✓ After deduplication: {len(df)} subjects")

    # Create target

    df['label_asd'] = (df['DX_GROUP'] == 1).astype(int)

    print(f"\n ASD: {df['label_asd'].sum()}")

    print(f" TDC: {(df['label_asd'] == 0).sum()}")

    print(f" Age range: {df['AGE_AT_SCAN'].min():.1f} -
{df['AGE_AT_SCAN'].max():.1f} years")

    return df
```

```
def select_features(df):

    """Select the 40 features used in the working model"""

    # These are the exact features from the working 95.96% model

    feature_list = [

        'ADI_R_SOCIAL_TOTAL_A', 'ADI_R_VERBAL_TOTAL_BV',

        'ADI_RRB_TOTAL_C',

        'ADI_R_ONSET_TOTAL_D', 'ADI_R_RSRCH_RELIABLE',

        'ADOS_STEREO_BEHAV',

        'ADOS_GOTHAM_SOCOAFFECT', 'ADOS_GOTHAM_RRB',

        'ADOS_GOTHAM_TOTAL',

        'ADOS_GOTHAM_SEVERITY', 'SRS_VERSION', 'SRS_RAW_TOTAL',

        'SRS_AWARENESS', 'SRS_COGNITION', 'SRS_COMMUNICATION',

        'SRS_MOTIVATION', 'SRS_MANNERISMS', 'SCQ_TOTAL', 'AQ_TOTAL',

        'OFF_STIMULANTS_AT_SCAN', 'VINELAND_RECEPTIVE_V_SCALED',

        'VINELAND_EXPRESSIVE_V_SCALED',

        'VINELAND_WRITTEN_V_SCALED',

        'VINELAND_COMMUNICATION_STANDARD',

        'VINELAND_PERSONAL_V_SCALED',

        'VINELAND_DOMESTIC_V_SCALED',

        'VINELAND_COMMUNITY_V_SCALED',

        'VINELAND_DAILYLVNG_STANDARD',

        'VINELAND_INTERPERSONAL_V_SCALED',

        'VINELAND_PLAY_V_SCALED', 'VINELAND_COPING_V_SCALED',

        'VINELAND_SOCIAL_STANDARD', 'VINELAND_SUM_SCORES',
```

```

'VINELAND_ABC_STANDARD', 'VINELAND_INFORMANT',
    'WISC_IV_SYM_SCALED', 'func_quality', 'func_mean_fd',
    'func_num_fd', 'func_perc_fd'
]

print("\n" + "="*80)

print("FEATURE SELECTION")

print("="*80)

# Keep only existing columns
available_features = [f for f in feature_list if f in df.columns]

print(f"✓ Using {len(available_features)} features")

# Extract features
X = df[available_features].copy()

# Fill missing values with median
X = X.fillna(X.median())

return X, available_features

def train_model(X, y, feature_names):
    """Train the XGBoost model matching the working configuration"""

    if not HAS_XGB:
        raise RuntimeError("XGBoost is required. Install with: pip install xgboost")

    print("\n" + "="*80)

```

```

print("TRAINING MODEL")

print("="*80)

# Split data with same random state as working model
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)

print(f"Train set: {len(y_train)} samples")
print(f"Test set: {len(y_test)} samples")
print(f" Train ASD/TDC: {y_train.sum()}/{(y_train==0).sum()}")
print(f" Test ASD/TDC: {y_test.sum()}/{(y_test==0).sum()}")

# Feature selection with SelectKBest
selector = SelectKBest(mutual_info_classif, k=min(40, X_train.shape[1]))
X_train_selected = selector.fit_transform(X_train, y_train)
X_test_selected = selector.transform(X_test)

print(f"\n✓ Selected {X_train_selected.shape[1]} features via SelectKBest")

# Scale features
scaler = RobustScaler()
X_train_scaled = scaler.fit_transform(X_train_selected)
X_test_scaled = scaler.transform(X_test_selected)

# Train XGBoost model
model = xgb.XGBClassifier(

```

```
n_estimators=300,
max_depth=6,
learning_rate=0.05,
subsample=0.8,
colsample_bytree=0.8,
reg_alpha=0.5,
reg_lambda=1.0,
scale_pos_weight=(len(y_train) - y_train.sum()) / (y_train.sum() + 1),
random_state=42,
verbosity=0
)

model.fit(X_train_scaled, y_train)

# Evaluate
y_pred = model.predict(X_test_scaled)
y_pred_proba = model.predict_proba(X_test_scaled)[:, 1]

acc = accuracy_score(y_test, y_pred)
prec = precision_score(y_test, y_pred, zero_division=0)
rec = recall_score(y_test, y_pred, zero_division=0)
f1 = f1_score(y_test, y_pred, zero_division=0)
roc = roc_auc_score(y_test, y_pred_proba)

print("\n" + "="*80)

print("TEST SET PERFORMANCE")

print("="*80)
```

```
print(f'Accuracy: {acc:.4f} ({acc*100:.2f}%)')

print(f'Precision: {prec:.4f}')

print(f'Recall: {rec:.4f}')

print(f'F1-Score: {f1:.4f}')

print(f'ROC-AUC: {roc:.4f}')


# Cross-validation

print("\n" + "="*80)

print("10-FOLD CROSS-VALIDATION")

print("="*80)


from sklearn.pipeline import Pipeline

pipeline = Pipeline([

    ('scaler', RobustScaler()),

    ('classifier', model)

])

cv_scores = cross_val_score(pipeline, X_train_selected, y_train,

                             cv=10, scoring='accuracy', n_jobs=1)

print(f'CV Accuracy: {cv_scores.mean():.4f} ± {cv_scores.std():.4f}')

print(f'Range: {cv_scores.min():.4f} - {cv_scores.max():.4f}')


# Confusion matrix

cm = confusion_matrix(y_test, y_pred)

print("\n" + "="*80)

print("CONFUSION MATRIX")
```

```
print("="*80)

print(f'TDC correctly classified: {cm[0,0]}')

print(f'TDC misclassified as ASD: {cm[0,1]}')

print(f'ASD correctly classified: {cm[1,1]}')

print(f'ASD misclassified as TDC: {cm[1,0]}')


print("\n" + "="*80)

print("CLASSIFICATION REPORT")

print("="*80)

print(classification_report(y_test, y_pred, target_names=['TDC', 'ASD']))


metrics = {

    'test_accuracy': acc,

    'test_precision': prec,

    'test_recall': rec,

    'test_f1': f1,

    'test_roc_auc': roc,

    'cv_accuracy_mean': cv_scores.mean(),

    'cv_accuracy_std': cv_scores.std(),

    'cv_accuracy_min': cv_scores.min(),

    'cv_accuracy_max': cv_scores.max(),

}

return model, selector, scaler, metrics


def main():

    print("\n" + "="*80)
```



```
print("ASD DETECTION MODEL - TRAINING SCRIPT")

print(f'Started: {datetime.now().strftime("%Y-%m-%d %H:%M:%S")}')

print("="*80)

# Load data

df = load_data()

# Select features

X, feature_names = select_features(df)

y = df['label_asd'].values

# Train model

model, selector, scaler, metrics = train_model(X, y, feature_names)

# Save artifacts

print("\n" + "="*80)

print("SAVING MODEL ARTIFACTS")

print("="*80)

model_artifacts = {

    'model': model,

    'selector': selector,

    'scaler': scaler,

    'feature_names': feature_names,

    'metrics': metrics

}
```

```
joblib.dump(model_artifacts, 'improved_asd_model_FIXED.pkl')

print("✓ improved_asd_model_FIXED.pkl")

# Save feature data

feature_df = df[feature_names + ['label_asd', 'SUB_ID']].copy()

feature_df.to_csv('improved_model_features_FIXED.csv', index=False)

print("✓ improved_model_features_FIXED.csv")

# Save metrics

pd.DataFrame([metrics]).to_csv('improved_model_metrics_FIXED.csv', index=False)

print("✓ improved_model_metrics_FIXED.csv")

print("\n" + "="*80)

print(f"COMPLETED: {datetime.now().strftime('%Y-%m-%d %H:%M:%S')}")

print(f"Final Test Accuracy: {metrics['test_accuracy']*100:.2f}%")

print("="*80 + "\n")

if __name__ == "__main__":

    main()
```

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